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Sentiment analysis of social media response on the Covid19 outbreak

The world has been grappling with COVID-19 disease ever since news about it and the positive cases for the disease thereof hit the stands. The impact of the outbreak has been so huge that it has been compared to dreaded epidemics and pandemics of the past like 'The Great Influenza' (Spanish flu of 1918), or the Black Death (form of bubonic plague). The scare around this outbreak has traversed across the globe affecting millions of people either through infection or through disruption, stress, worry, fear, disgust, and sadness (Montemurro, 2020). The virus strain, considered as a novel one, has so far killed more than two hundred thousand people and breached borders across the world (Lin et al., 2020). Experts around the world are researching upon the issue from a multitude of perspectives. Such a research pursuit also includes assessing how Artificial Intelligence (AI) could explain and predict any patterns caused by the novel Coronavirus (Qiu et al., 2020). At the same time, governments have implemented many measures like social distancing and isolation to prevent the spread of this virus. For netizens, social media have become a significant interface to share vital information and also a potent space of any misinformation for many users around the world ("Infodemic", 2020). The pandemic has been the most trending and talked about issue online ever since it was first reported in the last week of February 2020.

The proliferation of social media usage for articulation of opinions and feelings by the common public has created possibilities of analysing such sentiments about any dominant and prevalent discourse. In respect of Coronavirus, we analysed sentiments expressed globally over one of the most popular social media interfaces i.e. Twitter – a microblogging site. The tweets were retrieved from Twitter with two important hashtags related to the pandemic – #COVID-19 and #Coronavirus – in order to study the perspectives of the Twitter users about the Coronavirus disease. For the purpose of this analysis, 92,646 and 85,513 tweets were used with hashtags #COVID-19 and #Coronavirus respectively. Applying the sentiment analysis to these tweets, it was observed that most of the tweets i.e. 48,157 (51.97%) expressed positive sentiments, while 31,553 (34.05%) were neutral and rest of the tweets-amounting to 12,936 (13.96%) – accounted for negative sentiments in case of #COVID-19. The sentiment analysis of #Coronavirus indicated that the sentiments recorded were mostly neutral as seen in 35,296 (41.27%) tweets, followed by positive sentiments expressed in 34,989 (40.91%) tweets, and 15,228 (17.80%) comprised of negative sentiments. Thus, the findings reveal that the perception of Twitter users was mostly positive or neutral whenever they used any of these two hashtags while tweeting. Table 1 presents some examples of tweets that show how users have been reacting over Twitter. It is interesting to note that users expressed least negative sentiments for both the hashtags. It indicates that there has been less concern and involvement on part of people about the disease. One of the major factors regarding the prominence of positive sentiments over social media during this pandemic is that most of the people, despite being stressed and under lockdown, appreciated the efforts of their respective governments and frontline warriors like the health workers, police personnel, etc. The users appear hopeful about timely efforts and action by their governments and officials such as complete lockdown, social distancing, and practising proper hygiene measures such as washing hands frequently and using alcohol-based sanitizers would defeat the pandemic soon. Various measures and restrictions have been positively ascertained regardless of the challenges.

Overall, it is observed that the sentiments are mostly positive indicating that even though Twitter users are currently quarantined or staying at home, yet they are hopeful and experiencing a different unique socialization opportunity with family. Implementation of various strategies vis-à-vis work-from-home and learning new skills are part of this experience. In future, it would also be useful to see the trends on sentiments with temporal and geographical information for various strategies such as lockdown, social distancing etc., undertaken to combat the coronavirus disease. The role of social media analysis through various AI approaches could present fresh avenues for research that would assist even in fields like life sciences, social sciences and machine learning.

Table 1
Example tweets from the dataset.

Tweet	Sentiment
This is wonderful! Mayo Clinic hometown Hilton hotel shows its support for health care workers during this trying time.	Positive
Great news: Save-on-Foods employees will receive a $2/hour pay boost in response to the incredible work they are doing.	Positive
Let unite to fight against Covid 19. Listen to the government. Stay at home. Now is the best time to practice.	Positive
Erdo fan sends audio message to mobile users across Turkey in bid to raise awareness of #COVID19 among elderly.	Neutral
From Bergamo, Italy; of the worst hit hospitals by #COVID19 have a dear friend in Italy. She said it is hell.	Negative
With some of our students facing food insecurity, especially during this difficult time with #COVID19, please consider.	Negative

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